ALGORITHM

Algorithm for the Motor Current Safety Monitoring System

1. Start the program.
2. Display the project title: “MOTOR CURRENT SAFETY MONITORING SYSTEM”.
3. Ask the user to enter the number of motors to be checked.
4. Initialize the following variables:
   - Safe Motor = 0
   - Overloaded Motor = 0
5. Repeat the following steps for each motor:
   - Enter the motor name or ID.
   - Enter the rated current.
   - Enter the measured current.
6. Validate the inputs:
   - If the rated current is less than 0 OR
   - If the measured current is less than 0
     - Display “Invalid Current Value”.
     - Skip the calculation for that motor and continue to the next motor.
7. Calculate the current difference:
   - Formula: Current Difference = Measured Current – Rated Current
8. Compare the measured current with the rated current.
   - If Measured Current ≤ Rated Current
     - Display Safe Operation
     - Increase safe Motors by 1
   - Otherwise
     - Display Overload Warning
     - Increase overloaded Motor by 1.
9. Display the motor report showing:
   - Motor Name
   - Rated Current
   - Measured Current
   - Current Difference
   - Status
10. Save the report into motor_current_report.txt
11. After all motors have been processed, display
    - Total Motors
    - Safe Motors
    - Overloaded Motors
12. Display: “Report Successfully Saved.”
13. Stop the program.